

Matthew Cromer

Education

University of Connecticut – *B.S. in Mechanical Engineering - 3.6/4.0 GPA*

August 2023 - May 2027

- Dean's List Spring 2026 with a 3.9 GPA
- Pursuing a double concentration in Dynamic Systems and Control (Robotics) as well as Design and Manufacturing

Experience

Mechanical Engineering Intern, Marmon Utility – *Amherst, NH*

May 2026 – August 2026

- Designed, prototyped, and manufactured multiple mechanical clamping boots using 3D printing; optimized the final design to improve tensile load capacity by 1,300% while maintaining electrical insulation performance.
- Executed mechanical and electrical validation testing (tensile, compression, cantilever, and wet/dry flashover testing) while performing root-cause investigation and material inspection using Lumafield X-ray CT scanning.
- Gained hands-on exposure to automated material handling, 6-axis robotic arms, and injection molding by working alongside an automation engineer on manufacturing lines.

Landscape Laborer, O'Neill Landscaping – *Elliot, ME*

May 2024 – August 2025

- Promoted after one season to independently operate company vehicles and complete customer job sites based on demonstrated reliability, accountability, and sound judgment.

Crew Member, Chipotle – *Stratham, NH*

February 2022 – December 2023

- Completed management-track training based on performance, learning front- and back-of-house operations, inventory management, and team leadership while balancing being a full-time student-athlete.

Projects

Autonomous Robotic Car

- Built and programmed an Arduino-based autonomous 4WD robotic vehicle integrating ultrasonic sensors and an IMU to perform obstacle avoidance, line following, and PID-based navigation.
- Programmed line-following, obstacle-avoidance, and camera-based behaviors in Arduino IDE using C++; debugged wiring/sensor-mapping issues using a circuit schematic

V6 Car Engine Design & Simulation

- Modeled 50+ individual components and assembled a fully constrained, functioning V6 engine
- Applied GD&T principles and engineering tolerances to critical interfaces, ensuring manufacturable fits, assembly compatibility, and dynamic clearances.
- Created a motion study animating piston-crankshaft interaction and cylinder firing sequence to validate kinematic behavior

Drone Analysis

- Performed FEA on a quadcopter frame (SpaceClaim geometry, ANSYS Mechanical) under combined thrust, gravity, and drag loads, including mesh convergence verification
- Identified safety factors of 10–15 across the frame, indicating the design was significantly over-built and had clear opportunity for mass reduction without sacrificing structural integrity
- Pinpointed the propeller-to-hub junction as the critical stress location (1.5 MPa, still safety factor >10), guiding where any future lightweighting should focus

Skills

Software: SolidWorks CAD/CAM/PDM, ANSYS Workbench, SpaceClaim, Discovery, Technical Drawings (GD&T), MoldFlow, Simulink, LabVIEW, Visual Studio

Programming/Hardware: MATLAB, Python, C++, HTML (with some CSS & JavaScript), Microsoft Excel, Azure AI Foundry, RAG/LLM Integration, Arduino, Raspberry Pi.

Manufacturing: Additive manufacturing/3D Printing, Mechanical testing (tensile, bending, torque), Lean Principles, Failure Mode and Effects Analysis (FMEA), Injection Molding, PCB Soldering